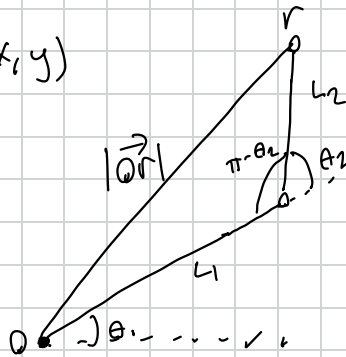


Derivation of analytical solution for θ_2

$$r = (x, y)$$



Law of cosines

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

$$|\vec{r}|^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(\pi - \theta_2)$$

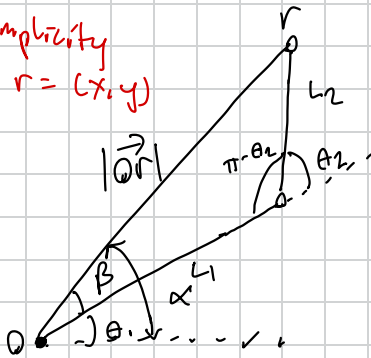
$$\cos(\pi - \theta) = -\cos(\theta)$$

$$|\vec{r}|^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos \theta_2$$

$$\cos(\theta_2) = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

Analytical solution for θ_1 given θ_2 :

For simplicity
 $\vec{r} = r = (x, y)$



Law of cosine:

$$L_2^2 = L_1^2 + |\vec{r}|^2 - 2L_1|\vec{r}| \cos(\beta)$$

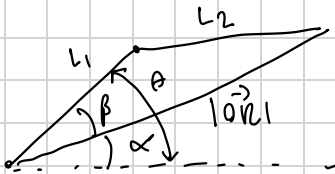
$$\cos(\beta) = \frac{L_1^2 + |\vec{r}|^2 - L_2^2}{2L_1|\vec{r}|}$$

$$\beta = \arccos\left(\frac{L_1^2 + |\vec{r}|^2 - L_2^2}{2L_1|\vec{r}|}\right)$$

$$\alpha = \text{atan2}(y, x)$$

$$\theta_1 = \alpha - \beta$$

$$\theta_1 = \alpha + \beta$$



Finding Jacobian for numerical solution. 2D

$$J(\theta_1, \theta_2) = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix} \quad f(\theta_1, \theta_2) = \begin{bmatrix} x(\theta_1, \theta_2) \\ y(\theta_1, \theta_2) \end{bmatrix}$$

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

$$\frac{\partial x}{\partial \theta_1} = -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2), \quad \frac{\partial x}{\partial \theta_2} = -L_2 \sin(\theta_1 + \theta_2)$$

$$\frac{\partial y}{\partial \theta_1} = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2), \quad \frac{\partial y}{\partial \theta_2} = L_2 \cos(\theta_1 + \theta_2)$$

Idea behind numerical solver:

$$\Delta p = \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad \Delta p = J \Delta \theta, \text{ maps joint change to end-effector change.}$$

$$\Delta \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad \text{error} = p_{\text{target}} - p_{\text{current}}$$

Since we know p_{target} , J and p_{current}

$$J \Delta \theta \approx \text{error}$$

$\Delta \theta \approx J^+ \text{error}$. we use the pseudoinverse

(Moore-Penrose inverse) to solve.